



The Scots College

HSC Mathematics Extension 2

Trial Examination

12th August 2011

Name: _____

General Instructions

- Working time : 3 hours + 5 minutes reading time.
- Write using blue or black pen
- Board approved calculators may be used (Non Graphic)
- All necessary working should be shown in every question
- Standard Integrals Table attached
- Answer each question on a SEPARATE answer booklet

TOTAL MARKS: 120

Attempt Questions 1 - 8
All questions are of equal value

WEIGHTING: 40 %

Question 1 (Marks 15) Use a SEPARATE writing booklet.

a) Evaluate

[2]

$$\int_0^{\frac{\pi}{4}} \frac{\sin x - \cos x}{\sin x + \cos x} dx$$

b) Find

[3]

$$\int \frac{dx}{\sqrt{x(2-x)}}$$

using the substitution $\sqrt{\frac{x}{2}} = \sin \theta$

c) Show that

[4]

$$\int_1^3 \frac{2x^2 - 3x + 11}{(x+1)(x^2 - 2x + 5)} dx = 2 \ln 2 + \frac{\pi}{8}$$

d) Find

[3]

$$\int x \ln(x+1) dx$$

e) Find

[3]

$$\int \frac{dx}{1 + \sin x + \cos x}$$

Question 2 (Marks 15) Use a SEPARATE writing booklet.

a) i) Find the square root of $-5 - 12i$. [2]

ii) Hence solve $z^2 - iz + 1 + 3i = 0$, expressing your answer in the form $a + ib$, where a and b are real numbers. [2]

b) i) Write $\frac{\sqrt{3} + i}{\sqrt{3} - i}$ in the modulus – argument form. [2]

ii) Hence express $\left(\frac{\sqrt{3} + i}{\sqrt{3} - i}\right)^{10}$ in the form $x + iy$, where x and y are both real. [2]

c) i) Sketch on the Argand diagram the locus of the complex number z , which satisfies the condition [2]

$$\arg\left(\frac{z-2}{z-2i}\right) = \frac{\pi}{2}$$

(ii) Hence, or otherwise, find the complex number z (in the form $a + ib$, where a and b are both real) which has the maximum value of $|z|$. [1]

d) Let $z = \cos\frac{\pi}{5} + i \sin\frac{\pi}{5}$

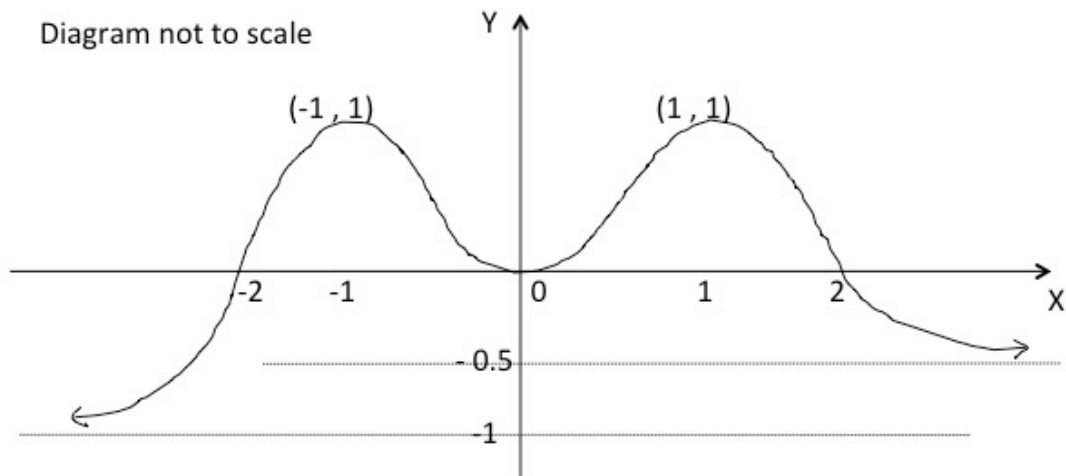
i) Show that $1 - z + z^2 - z^3 + z^4 = 0$ [2]

ii) Show that $(1 - z)(1 + z^2)(1 - z^3)(1 + z^4) = 1$ [2]

Question 3 (Marks 15) Use a SEPARATE writing booklet.

a) The graph of $y = f(x)$ is given below.

[10]



Using the graph of $y = f(x)$, sketch on separate axes, the graphs of

- i. $|y| = |f(x)|$
- ii. $y^2 = f(x)$
- iii. $y = \frac{1}{f(x)}$
- iv. $y = e^{f(x)}$
- v. $y = \sin^{-1} f(x)$

b) (i) Show that if $I_n = \int_0^1 x^n \sqrt{1-x^2} dx$, then

[3]

$$I_n = \frac{n-1}{n+2} I_{n-2}$$

(ii) Hence evaluate $\int_0^1 x^4 \sqrt{1-x^2} dx$

[2]

Question 4 (Marks 15) Use a SEPARATE writing booklet.

- a) A hyperbola has the asymptotes $y = x$ and $y = -x$, and it passes through the point $(5,4)$. Find [7]

- i. the equation of the hyperbola
- ii. its eccentricity
- iii. the length of the principal axis
- iv. the coordinates of the foci
- v. equation of the directrices
- vi. the length of the latus rectum

- b) The point $P \left(cp, \frac{c}{p} \right)$, lies on the hyperbola $xy = c^2$. The tangent at P meets the x – axis at A and the y – axis at B . The normal to the hyperbola at P meets the line $y = x$ at the point C . [8]

- i. Show that the equation of the tangent at P is $x + p^2y = 2cp$.
- ii. Find the coordinates of A and B .
- iii. Find the equation of the normal at P .
- iv. Show that the x – coordinate of the point C is given by $x = \frac{c}{p} (p^2 + 1)$.
- v. Prove that $\triangle ABC$ is an isosceles triangle.

Question 5 (Marks 15) Use a SEPARATE writing booklet.

a) $P (a \cos \theta, b \sin \theta)$ is a variable point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

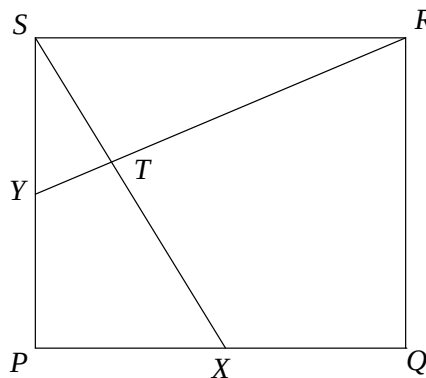
i. Show that the equation of the tangent to the ellipse at P is $bx \cos \theta + ay \sin \theta = ab$. [2]

ii. Deduce the equation of the normal to the ellipse at P . [2]

iii. Find the coordinates of X and Y , the points where the tangent and the normal respectively, meet the y – axis. [2]

iv. Show that the circle with XY as the diameter, passes through the foci of the ellipse.. [3]

b) $PQRS$ is a square. X and Y are mid-points of PQ and PS respectively. SX and RY intersect at the point T . [6]



i. Prove that $QRTX$ is a cyclic quadrilateral.

ii. Hence prove that $QT = QR$

Question 6 (Marks 15) Use a SEPARATE writing booklet.

- a) A solid is formed by rotating the circle $x^2 - 2ax + y^2 = 0$ about the line $x = 3a$. [5]

Find the volume of the solid generated by taking slices perpendicular to the axis of rotation.

- b) The base of a certain solid is the region between the curve $y = \frac{x^3}{4}$, $0 \leq x \leq 2$, and the line $y = x$. [6]

Each plane section of the solid perpendicular to the x-axis is a parabola whose chord lies on the base of the solid, with one end point A on the line $y = x$ and the other end point B on the curve $y = \frac{x^3}{4}$. The axis of the parabola is vertical and passing through the mid-point of AB and the maximum height of the parabola, from the base, is equal to the length of the chord AB.

By first finding the area of a slice taken perpendicular to the x – axis, find the volume of the solid.

- c) A sequence $u_1, u_2, u_3, \dots$ is defined by the relation [4]

$$u_n = u_{n-1} + 6u_{n-2}, \text{ for } n \geq 3.$$

Given that $u_1 = 1$ and $u_2 = -12$, prove by using mathematical induction

$$u_n = -6[(-2)^{n-2} + 3^{n-2}], \text{ for all positive integers } n.$$

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Question 7 (Marks 15) Use a SEPARATE writing booklet.

- a) i. By using De Moivre's Theorem or otherwise, prove that [3]

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}.$$

- ii. Using part (i) solve the equation [4]

$$x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$$

and hence find the value of

$$\tan \frac{\pi}{16} \times \tan \frac{3\pi}{16} \times \tan \frac{5\pi}{16} \times \tan \frac{7\pi}{16}$$

- b) It is given that the product of two of the roots of the equation [4]
 $x^4 + x^3 - 16x^2 - 4x + 48 = 0$, is equal to 6.

Show that the equation can be written in the form
 $(x^2 + ax + b)(x^2 + cx + d) = 0$, where a, b, c and d are integers.

Hence or otherwise solve the equation.

- c) i. Find all the values of m for which the polynomial $3x^4 - 4x^3 + m = 0$ [4]
has no real roots.

- ii. Determine the real roots of the polynomial when $m = 1$

Question 8 (Marks 15) Use a SEPARATE writing booklet.

- a) A particle P of mass m kg is projected vertically upwards from the ground, [8]
with an initial velocity of u m/s, in a medium of resistance mkv^2 , where k is a
positive constant and v is the velocity of the particle.

- i. Show that the maximum height H , from the ground, attained by the
particle P is given by $H = \frac{1}{2k} \ln \left(1 + \frac{ku^2}{g} \right)$, where g is the acceleration
due to gravity.

- ii. At the same time that P is projected upwards, another particle, Q , of
equal mass, initially at rest, is allowed to fall downwards in the same
medium, from a height of H metres from the ground, along the same
vertical path as P . Show that at the time of collision of P and Q ,

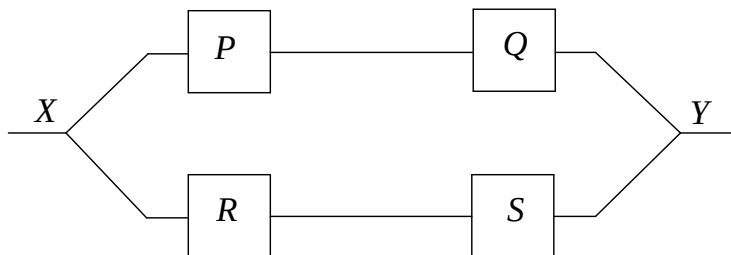
$$\frac{1}{v_2^2} - \frac{1}{v_1^2} = \frac{1}{V^2},$$

where v_1 and v_2 are the velocities of particles P and Q respectively, at
the time of collision, and $V = \sqrt{\frac{g}{k}}$.

- b) Find the stationary points, stating their nature, for the curve [3]
 $x^2 + y^2 = xy + 3$

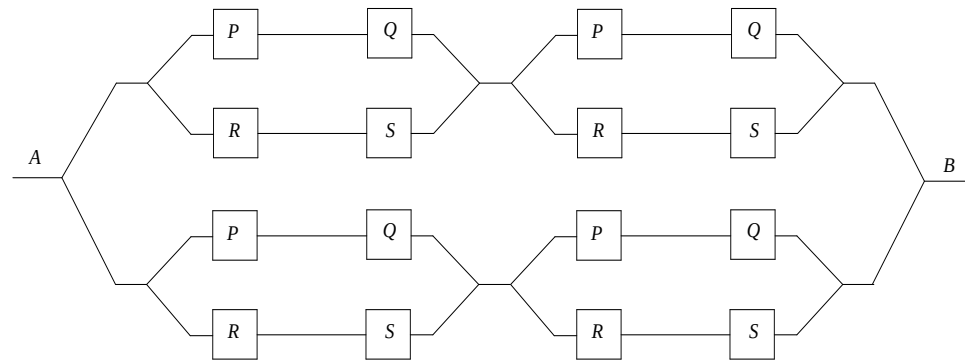
- c) i. An electrical circuit has four bulbs P , Q , R and S placed as shown in the [4]
diagram. The probability of each bulb being defective, independently, is
given by p . Current can flow from X to Y through either or both of the
branches of the electrical circuit. However, no current will flow through a
branch that has at least one defective bulb.

Show that the probability that the current *does not* flow from X to Y is
 $(2p - p^2)^2$



Question 8 continued.....

- ii. In the household there are four such circuits in one connection. Find the probability that current does not flow from A to B .



End of Assessment

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Mathematics Ext 2 Trials 2011Question No. 1

$$(a) \int_0^{\pi/4} \frac{\sin x - \cos x}{\sin x + \cos x} dx$$

$$= - \int_0^{\pi/4} \frac{\cos x - \sin x}{\sin x + \cos x} dx$$

$$= - \left[\ln [\sin x + \cos x] \right]_0^{\pi/4}$$

$$= - \left[\ln \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) - \ln (0+1) \right]$$

$$= - \ln \frac{2}{\sqrt{2}}$$

$$= - \ln \sqrt{2}$$

$$= - \frac{1}{2} \ln 2$$

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ANSWER SHEET

Name: _____

Teacher: _____

Question No. 1.

$$\int \frac{dx}{\sqrt{x(2-x)}}$$

$$\sqrt{\frac{x}{2}} = \sin \theta$$

$$x = 2 \sin^2 \theta$$

$$dx = 4 \sin \theta \cos \theta d\theta$$

$$= \int \frac{4 \sin \theta \cos \theta d\theta}{\sqrt{2 \sin^2 \theta \cdot 2 \cos^2 \theta}}$$

$$= \int \frac{4 \sin \theta \cos \theta}{2 \sin \theta \cos \theta} d\theta$$

$$= \int 2 d\theta$$

$$= 2\theta + C$$

$$= 2 \sin^{-1} \sqrt{\frac{x}{2}} + C$$

$$(C) \int_1^3 \frac{2x^2 - 3x + 11}{(x+1)(x^2 - 2x + 5)} dx = 2\ln 2 + \frac{\pi}{8}$$

$$\frac{2x^2 - 3x + 11}{(x+1)(x^2 - 2x + 5)} = \frac{A}{x+1} + \frac{Bx+C}{x^2 - 2x + 5}$$

$$= \frac{A(x^2 - 2x + 5) + (Bx+C)(x+1)}{(x+1)(x^2 - 2x + 5)}$$

$$2x^2 - 3x + 11 = A(x^2 - 2x + 5) + (Bx+C)(x+1)$$

$$x = -1$$

$$2 + 3 + 11 = A(1 + 2 + 5)$$

$$16 = 8A \Rightarrow A = 2$$

$$x = 0$$

$$11 = 2(5) + C(1)$$

$$\therefore C = 1$$

Equating coeff of x^2

$$2 = 2 + B \therefore B = 0$$

$$\therefore \int_1^3 \frac{2x^2 - 3x + 11}{(x+1)(x^2 - 2x + 5)} dx = \int_1^3 \left[\frac{2}{x+1} + \frac{1}{(x-1)^2 + 4} \right] dx$$

$$= \left[2\ln(x+1) \right]_1^3 + \left[\frac{1}{2} \tan^{-1} \frac{x-1}{2} \right]_1^3$$

$$= (2\ln 4 - 2\ln 2) + \left[\frac{1}{2} \tan^{-1} 1 - 2 \tan^{-1} 0 \right]$$

$$= 2\ln 2 + \frac{1}{2} \cdot \frac{\pi}{4}$$

$$= 2\ln 2 + \frac{\pi}{8}$$



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Question No. _____

$$(d) \int x \ln(x+1) dx \quad \begin{array}{l} u = x \\ \frac{du}{dx} = 1 \end{array} \quad \begin{array}{l} v = \ln(x+1) \\ v' = \frac{1}{x+1} \end{array}$$

$$= \frac{x^2}{2} \ln(x+1) - \int \frac{x^2}{2(x+1)} dx$$

$$= \frac{x^2}{2} \ln(x+1) - \frac{1}{2} \int \frac{x^2 - 1 + 1}{x+1} dx$$

$$= \frac{x^2}{2} \ln(x+1) - \frac{1}{2} \int \left(x - 1 + \frac{1}{x+1} \right) dx$$

$$= \frac{x^2}{2} \ln(x+1) - \frac{1}{2} \left[\left(\frac{x^2}{2} - x \right) + \ln(x+1) \right] + C$$



ANSWER SHEET

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Question No. _____

$$(e) \int \frac{dx}{1 + \sin^2 x + \cos^2 x}$$

$$\text{let } t = \tan \frac{x}{2}$$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

$$= \frac{1}{2} (1 + t^2) dx$$

$$dx = \frac{2dt}{1+t^2}$$

$$= \int \frac{2dt / (1+t^2)}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}$$

$$= \int \frac{2dt / (1+t^2)}{\frac{1+t^2+2t+1-t^2}{1+t^2}}$$

$$= \int \frac{2dt}{2+2t}$$

$$= \int \frac{dt}{1+t}$$

$$= \ln(1+t) + C$$

$$= \ln\left(1 + \tan \frac{x}{2}\right) + C$$



ANSWER SHEET

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Question No. 2

$$(Q) \quad (i) \quad \text{let } \sqrt{-5-12i} = x+iy$$

$$x^2 + 2ixy + y^2 = -5 - 12i$$

$$x^2 - y^2 = -5$$

$$2xy = -12 \Rightarrow xy = -6$$
$$y = -\frac{6}{x}$$

$$x^2 - \frac{36}{x^2} = -5$$

$$x^4 + 5x^2 - 36 = 0$$

$$(x^2 + 9)(x^2 - 4) = 0$$

$$x^2 = -9 \quad \text{or} \quad x^2 = 4$$

$$(\text{not}) \quad \therefore x = 2, y = -3$$

$$\text{or } x = -2, y = 3$$

$\therefore$ Square roots of

$$-5-12i \text{ are } \underline{2-3i} \text{ or } \underline{-2+3i}$$



ANSWER SHEET

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Teacher: _____

Question No. 2

$$(a) (ii) \quad z^2 - iz + 1 + 3i = 0$$

$$z = \frac{i \pm \sqrt{(-i)^2 - 4(1)(1+3i)}}{2(1)}$$

$$= \frac{i \pm \sqrt{-1 - 4 - 12i}}{2}$$

$$= \frac{i \pm \sqrt{-5 - 12i}}{2}$$

$$= \frac{i + 2 - 3i}{2} \text{ or } \frac{i - 2 + 3i}{2}$$

$$= \frac{2 - 2i}{2} \text{ or } \frac{-2 + 4i}{2}$$

$$= \underline{1 - i} \text{ or } \underline{-1 + 2i}$$



ANSWER SHEET

Name: _____

Teacher: _____

Question No. 2.

$$(b) (i) \frac{\sqrt{3} + i}{\sqrt{3} - i}$$

$$\sqrt{3} + i = 2(\cos \pi/6 + i \sin \pi/6)$$

$$\sqrt{3} - i = 2(\cos(-\pi/6) + i \sin(-\pi/6))$$

$$\begin{aligned} \frac{\sqrt{3} + i}{\sqrt{3} - i} &= \cos(\pi/6 + \pi/6) + i \sin(\pi/6 + \pi/6) \\ &= \cos \pi/3 + i \sin \pi/3 \end{aligned}$$

$$\begin{aligned} (ii) \left(\frac{\sqrt{3} + i}{\sqrt{3} - i} \right)^{10} &= \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)^{10} \\ &= \cos \frac{10\pi}{3} + i \sin \frac{10\pi}{3} \\ &= -\cos(\pi/3) + i \sin(\pi/3) \\ &= \underline{\underline{-\frac{1}{2} + \frac{\sqrt{3}}{2}i}} \end{aligned}$$



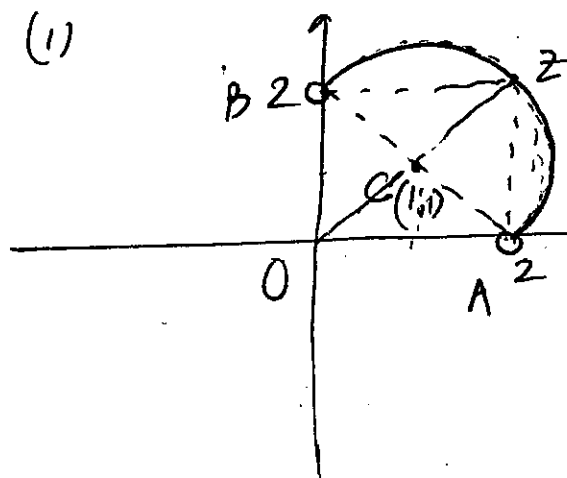
ANSWER SHEET

Name: _____

Teacher: _____

Question No. 2

(C) (i)



Locus is the semi-circle with AB as Diameter.

radius: $\frac{1}{2} \sqrt{2^2 + 2^2} = \frac{1}{2} \sqrt{8} = \sqrt{2}$

Equation of circle.

$$(x-1)^2 + (y-1)^2 = 2$$

(ii) Max. value of z lies along the line

$$y = x.$$

$$\therefore |z| = \sqrt{2} + \sqrt{2} = 2\sqrt{2}$$

$$\underline{\underline{z = 2 + 2i}}$$



ANSWER SHEET

Name: _____

Teacher: _____

Question No. 2

$$(d) \quad z = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}.$$

$$z^5 = \cos \frac{5\pi}{5} + i \sin \frac{5\pi}{5}$$

$$= -1 + 0$$

$$= -1$$

$$(i) \quad 1 - z + z^2 - z^3 + z^4$$

$$r = -2, \quad n = 5$$

$$S_5 = \frac{1 - (-2)^5}{1 - (-2)}$$

$$= \frac{1 + 2^5}{1 + 2} \quad (z \neq -1)$$

$$= \frac{0}{1 + 2} \quad (\because z^5 = -1)$$

$$(ii) \quad (1 - z)(1 + z^2)(1 - z^3)(1 + z^4)$$

$$= (1 - z + z^2 - z^3)(1 - z^3 + z^4 - z^7)$$

$$= (-z^4)(1 - z^3 + z^4 + z^2) \quad (\because 1 - z + z^2 - z^3 + z^4 = 0 \text{ and } z^7 = z^5 \cdot z^2 = -z^2)$$

$$= (-z^4)(z) \quad (\because 1 - z + z^2 - z^3 + z^4 = 0)$$

$$= -z^5 = 1 \text{ as required.}$$



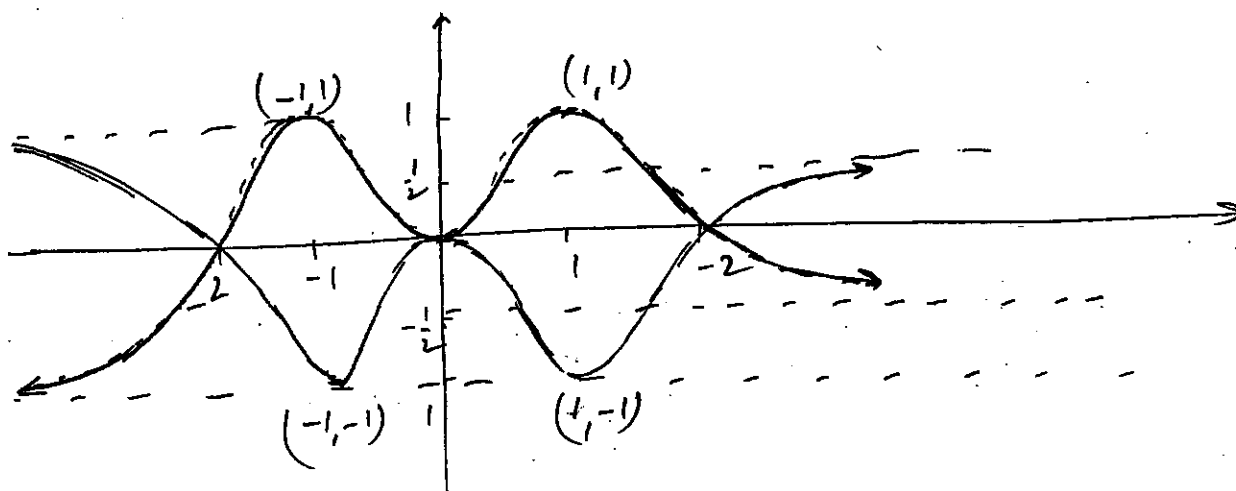
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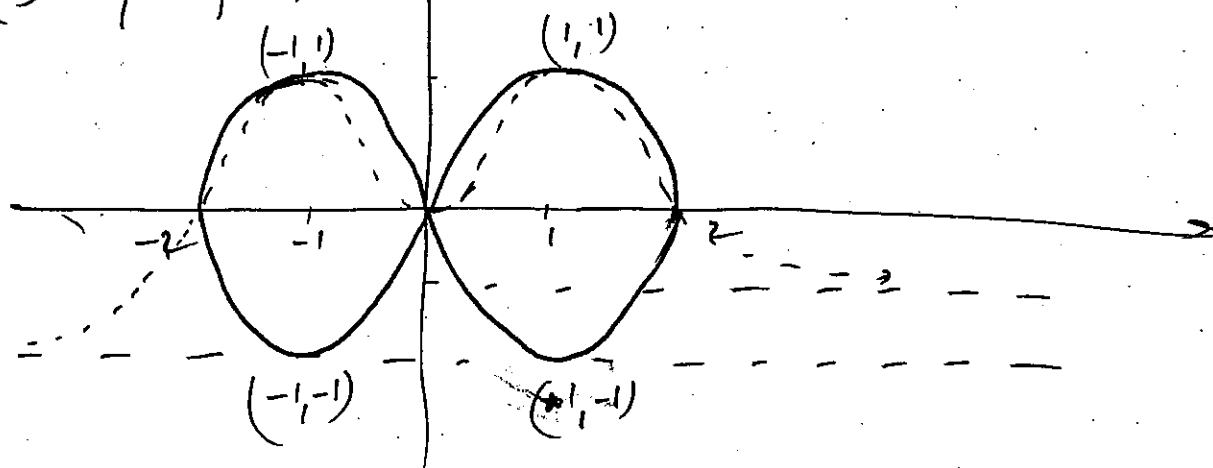
Teacher: _____

Question No. 3

(a) (i) $|y| = |f(x)|$



(ii) $y^2 = f(x)$





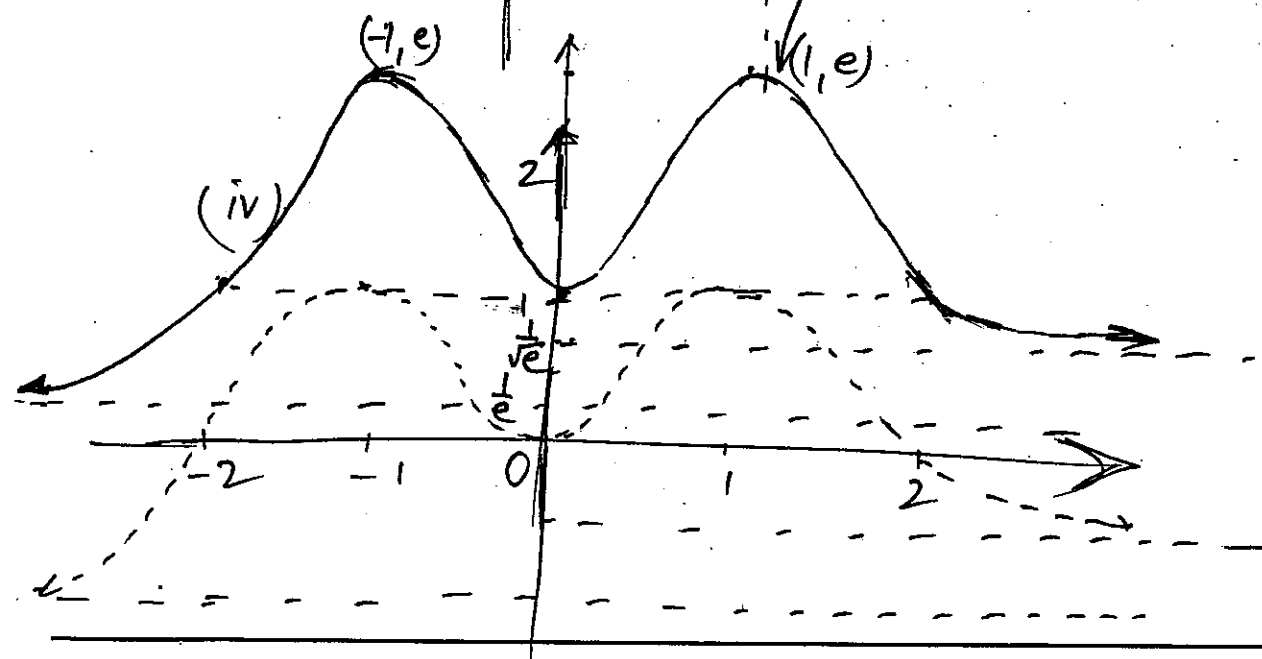
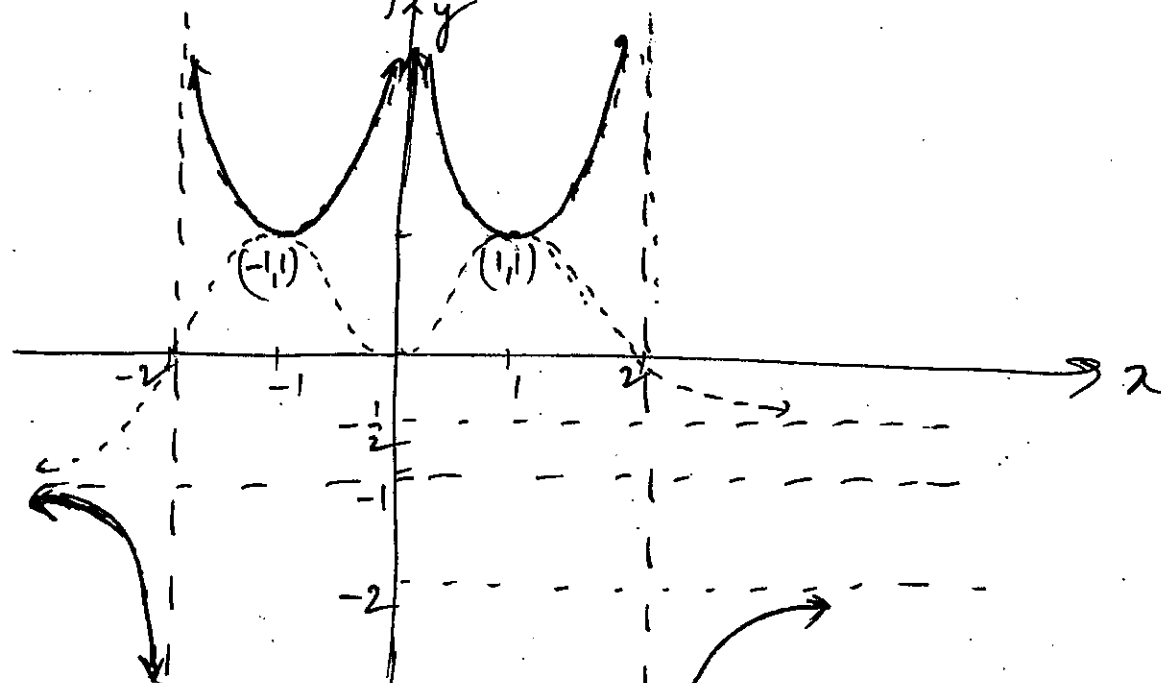
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Question No. 3

(Q) (iii) $y = \frac{1}{f(x)}$





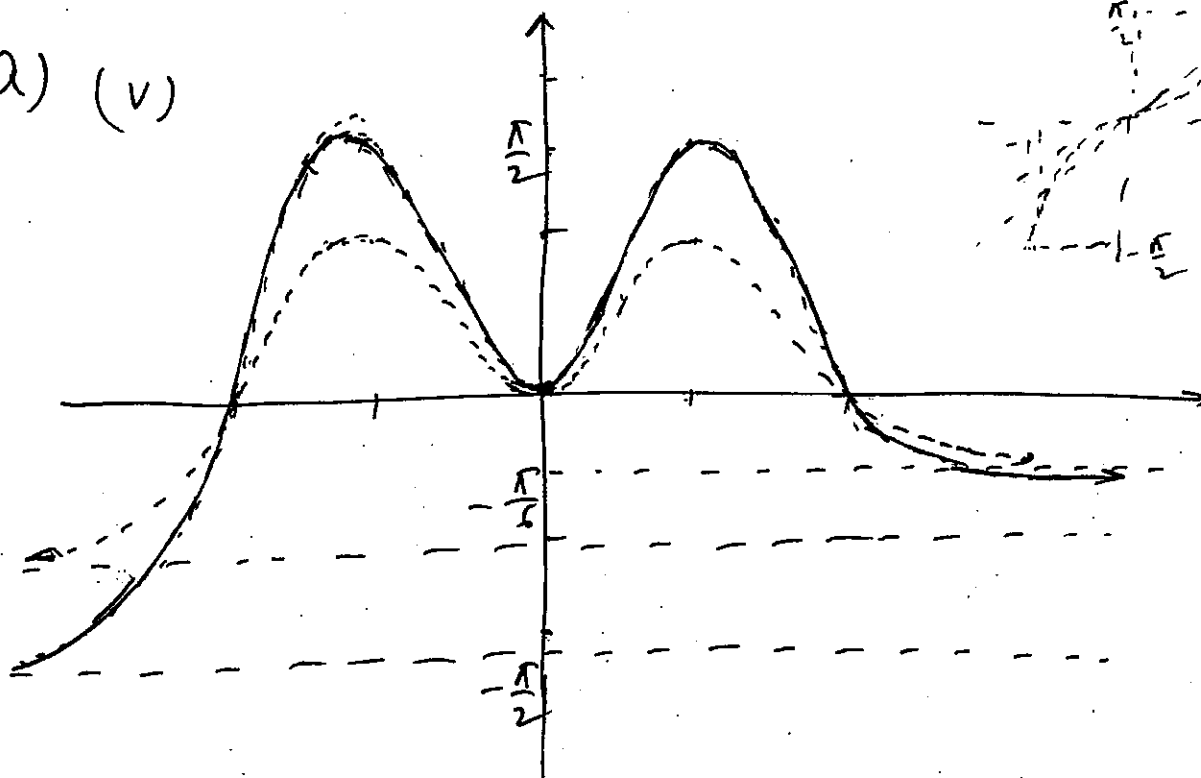
ANSWER SHEET

Name: _____

Teacher: _____

Question No. 3.

(a) (v)





ANSWER SHEET

Name: _____

Teacher: _____

Question No. 35

$$(b) \quad (1) \quad I_n = \int_0^1 x^n \sqrt{1-x^2} dx$$

$$u = x^{n-1} \quad v' = x \sqrt{1-x^2}$$

$$= \left[\frac{x^{n-1}}{3} (1-x^2)^{3/2} \right]_0^1 + \int_0^1 \frac{(n-1)x^{n-2}}{3} (1-x^2)^{3/2} dx$$
$$u' = (n-1)x^{n-2} \quad v = -\frac{1}{2 \cdot \frac{3}{2}} (1-x^2)^{3/2} = -\frac{1}{3} (1-x^2)^{3/2}$$

$$= (0-0) + \frac{n-1}{3} \int_0^1 x^{n-2} (1-x^2) (1-x^2)^{1/2} dx$$

$$= \frac{n-1}{3} \int_0^1 (x^{n-2} \sqrt{1-x^2} - x^n \sqrt{1-x^2}) dx$$

$$= \frac{n-1}{3} (I_{n-2} - I_n)$$

$$3 I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$3 I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$(n+2) I_n = (n-1) I_{n-2}$$

$$\therefore I_n = \frac{n-1}{n+2} I_{n-2}$$



ANSWER SHEET

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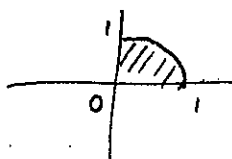
Question No. 3

$$(b) (ii) I_4 = \int_0^1 x^4 \sqrt{1-x^2} dx$$

$$= \frac{3}{5} I_2$$

$$= \frac{3}{5} \left[\frac{1}{3} I_0 \right]$$

$$I_0 = \int_0^1 \sqrt{1-x^2} dx$$



$$= \frac{\pi}{4} (1)^2 = \frac{\pi}{4}$$

$$\therefore \int_0^1 x^4 \sqrt{1-x^2} dx = \frac{3}{5} \times \frac{1}{3} \times \frac{\pi}{4}$$
$$= \underline{\underline{\frac{\pi}{20}}}$$



ANSWER SHEET

Name: _____

Teacher: _____

Question No. 4

(Q) Asymptotes are $y = x$ and $y = -x$

(i) $\therefore$ Rectangular hyperbola

$$x^2 - y^2 = a^2$$

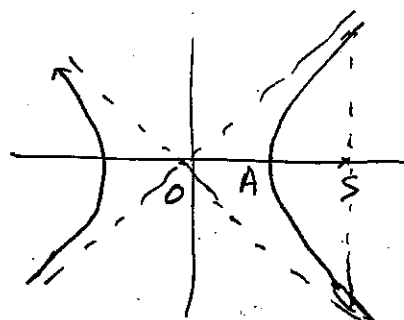
Sub (5, 4)

$$25 - 16 = a^2$$

$$a^2 = 9$$

$$a = 3$$

Equation $x^2 - y^2 = 9$



(ii) Eccentricity $= \sqrt{2}$

(iii) Length of principal axis $= 6$

(iv) Focus $= (\pm 3\sqrt{2}, 0)$

(v) Equation of director $x = \pm \frac{a}{e}$

$$x = \pm \frac{3}{\sqrt{2}}$$

(vi) Length of Latus Rectum: $2b^2/a = 2a^2/a$
 $= 2a = \underline{\underline{6}}$



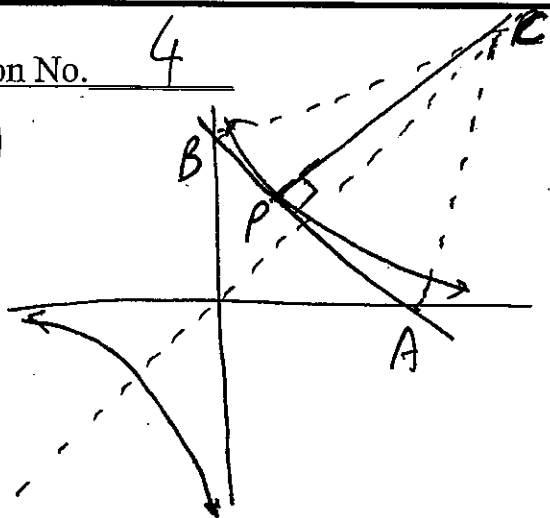
ANSWER SHEET

Name: _____

Teacher: _____

Question No. 4

(b)



(i) $y = \frac{c}{x}$

$$\frac{dy}{dx} = -\frac{c}{x^2}$$

at $x = cp$

$$\frac{dy}{dx} = -\frac{c}{c^2 p^2}$$

$$= -\frac{1}{p^2}$$

Equation of tangent

$$y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$$

$$p^2 y - cp = -x + cp$$

$$\text{or } \underline{x + p^2 y = 2cp}$$

(ii) $y = 0, x = 2cp \therefore A(2cp, 0)$

$$x = 0, y = \frac{2cp}{p^2} = \frac{2c}{p} \therefore B(0, \frac{2c}{p})$$



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Question No. 4

$$(b) (iii) m_N = p^2$$

Equation of normal

$$y - \frac{c}{p} = p^2(x - 2cp)$$

$$py - c = p^3x - 2cp^4$$

$$\text{or } p^3x - py = 2cp^4 - c$$

$$(iv) y = x$$

$$\therefore p^3x - px = c(p^4 - 1)$$

$$px(p^2 - 1) = c(p^2 - 1)(p^2 + 1)$$

$$\therefore x = \frac{c}{p}(p^2 + 1) \Rightarrow \left(cp + \frac{c}{p}\right)$$

$$(v) AC^2 = \left(cp + \frac{c}{p} - 2cp\right)^2 + \left(cp + \frac{c}{p} - \frac{2c}{p}\right)^2$$
$$= \left(\frac{c}{p} - cp\right)^2 + \left(cp + \frac{c}{p}\right)^2$$
$$= 2\left(\frac{c^2}{p^2} + c^2p^2\right)$$

$$BC^2 = \left(cp + \frac{c}{p} - 0\right)^2 + \left(cp + \frac{c}{p} - \frac{2c}{p}\right)^2 = \left(cp + \frac{c}{p}\right)^2 + \left(cp - \frac{c}{p}\right)^2$$
$$= 2\left(c^2p^2 + \frac{c^2}{p^2}\right)$$

$$\therefore AC = BC$$



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Question No. 5

$$(a) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad p(a \cos \theta, b \sin \theta)$$

(1) Differentiating w.r.t. x

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

When $x = a \cos \theta$, $y = b \sin \theta$

$$\frac{dy}{dx} = -\frac{b^2 \cdot a \cos \theta}{a^2 \cdot b \sin \theta} = -\frac{b \cos \theta}{a \sin \theta}$$

Equation of tangent

$$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$$

$$bx \cos \theta + ay \sin \theta = ab (\cos^2 \theta + \sin^2 \theta)$$

$$\underline{bx \cos \theta + ay \sin \theta = ab}$$



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Question No. 5

$$(a) (ii) M_N = \frac{a \sin \theta}{b \cos \theta}$$

Equation of normal

$$y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$$

$$by \cos \theta - b^2 \sin \theta \cos \theta = ax \sin \theta - a^2 \sin \theta \cos \theta$$

$$ax \sin \theta - by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

(iii) For X:

$$y=0 \quad bx \cos \theta = ab$$

$$\boxed{x = \frac{a}{\cos \theta}}$$

$$x=0, \quad ay \sin \theta = ab$$

$$y = \frac{b}{a \sin \theta}$$

$$X(0, \frac{b}{a \sin \theta})$$

For Y

$$y=0 \quad ax \sin \theta = (a^2 - b^2) \sin \theta \cos \theta$$

$$x = \frac{a^2 - b^2}{a} \cos \theta$$

$$-by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

$$y = -\frac{a^2 - b^2}{b} \sin \theta$$

$$Y(0, -\frac{a^2 - b^2}{b} \sin \theta)$$



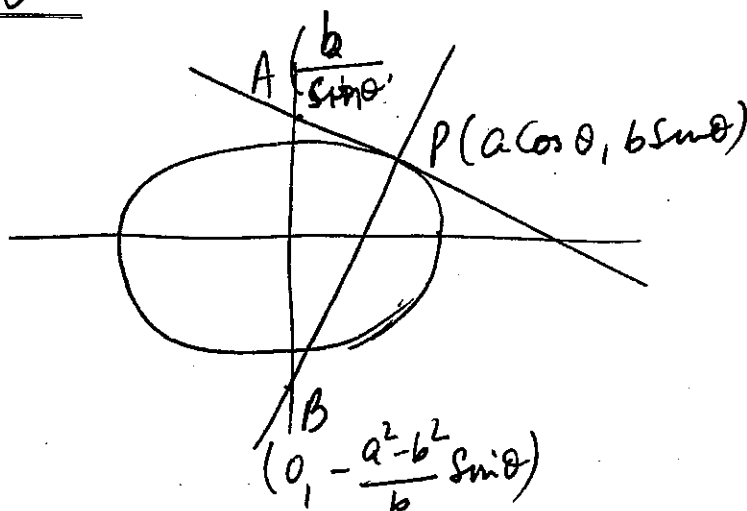
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Question No. 5

(2) (iv)



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$b^2 = a^2(1-e^2)$$

$$a^2 e^2 = a^2 - b^2$$

$$AB = \frac{b}{\sin \theta} + \frac{a^2 - b^2}{b} \sin \theta$$

$$= \frac{b^2 + (a^2 - b^2) \sin^2 \theta}{b \sin \theta}$$

$$= \frac{b^2 + a^2 e^2 \sin^2 \theta}{b \sin \theta}$$

radius: $\frac{b^2 + a^2 e^2 \sin^2 \theta}{2b \sin \theta}$

Mid pt of AB

$$x = 0$$

$$y = \left(\frac{b}{\sin \theta} - \frac{a^2 - b^2}{b} \sin \theta \right) \div 2$$

$$= \frac{b^2 - a^2 e^2 \sin^2 \theta}{2b \sin \theta}$$

$\therefore$ Equation of circle:

$$x^2 + \left(y - \frac{b^2 - a^2 e^2 \sin^2 \theta}{2b \sin \theta} \right)^2 = \left(\frac{b^2 + a^2 e^2 \sin^2 \theta}{2b \sin \theta} \right)^2$$

Sub $x = \pm ae, y = 0$

$$\text{LHS: } a^2 e^2 + \left(\frac{b^2 - a^2 e^2 \sin^2 \theta}{2b \sin \theta} \right)^2 = \frac{4a^2 b^2 e^2 \sin^2 \theta + (b^2 - a^2 e^2 \sin^2 \theta)^2}{4b^2 \sin^2 \theta}$$

$$= \left(\frac{b^2 + a^2 e^2 \sin^2 \theta}{2b \sin \theta} \right)^2$$

$\therefore$ Circle passes through S & S'.



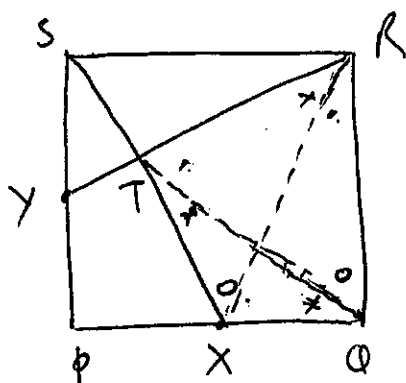
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Question No. 5

(b)



(i) In ΔPXS and ΔYRS

$PX = SY$ (half of equal sides of a square)

$PS = SR$ (equal sides of square)

$\angle SPX = \angle SRY = 90^\circ$ ($\because PQRS$ is a square)

$\therefore \Delta PXS \cong \Delta YRS$ (SAS)

$\therefore \angle PSX = \angle SRY$ (corresponding $\angle$'s of congruent Δ 's)

In ΔSYT and ΔRYT

$\angle SYT = \angle RYT$ (common angle)

$\angle YST = \angle SRY$ (proven above)

$\therefore \angle STY = \angle YSR$ (angle sum of Δ)

$\therefore \angle STY = 90^\circ$ ($\because \angle YSR = 90^\circ$, internal angle of square)

$\angle STY = \angle RQX = 90^\circ$

$\therefore \underline{QRTX \text{ is a cyclic quadrilateral.}}$



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Question No. 5

(b) (ii) Join XR

In $\triangle PSX$ and $\triangle QRX$

$PX = QX$ (X is mid pt of PQ)

$PS = QR$ (equal sides of square)

$\angle SPX = \angle RQX = 90^\circ$

$\therefore \triangle SPX \equiv \triangle QRX$ (SAS)

$\therefore \angle PXS = \angle QXR$ (corresponding $\angle$'s)

$\angle PXS = \angle QRT$ ($\angle$ ext. angle of cyclic quadrilateral equal to opp. interior $\angle$)

$\angle QXR = \angle QTR$ (angles in same segment)

$\therefore \angle QRT = \angle QTR$

$\therefore QR = QT$ (equal sides opposite to equal angles of a $\triangle QRT$)



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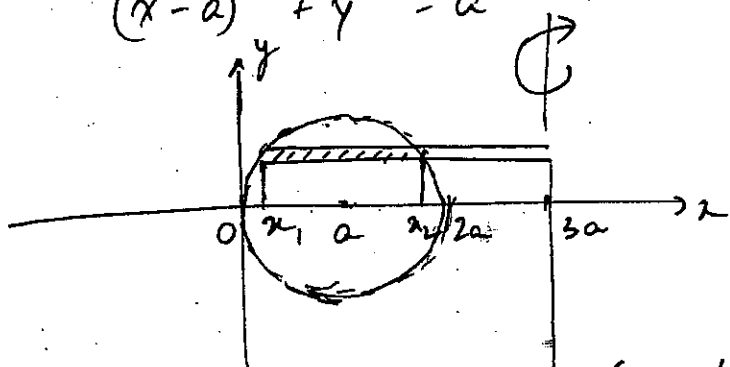
Teacher: _____

Question No. 6

$$(a) \quad x^2 - 2ax + y^2 = 0$$

$$\text{or } x^2 - 2ax + a^2 + y^2 = a^2$$

$$(x-a)^2 + y^2 = a^2$$



$$(x-a)^2 = a^2 - y^2$$

$$x-a = \pm \sqrt{a^2 - y^2}$$

$$x = a \pm \sqrt{a^2 - y^2}$$

$$x_2 + x_1 = 2a$$

$$x_2 - x_1 = 2\sqrt{a^2 - y^2}$$

Let a section of thickness δy be taken $\perp$ to $x = 3a$.
Volume of the section when rotated about $x = 3a$ is

$$\delta V = \pi \left[(3a - x_1)^2 - (3a - x_2)^2 \right] \delta y$$

$$= \pi \left[(6a - x_1 - x_2)(-x_1 + x_2) \right] \delta y$$

$$= \pi (6a - 2a)(2\sqrt{a^2 - y^2}) \delta y$$

$$= 8a\pi \sqrt{a^2 - y^2} \delta y$$

$$\therefore V = \int_{-a}^a 8a\pi \sqrt{a^2 - y^2} dy$$

$$= 8a\pi \int_{-a}^a \sqrt{a^2 - y^2} dy$$

$$= 8a\pi \cdot \frac{\pi a^2}{2} = \underline{\underline{4a^3\pi^2 u^3}}$$



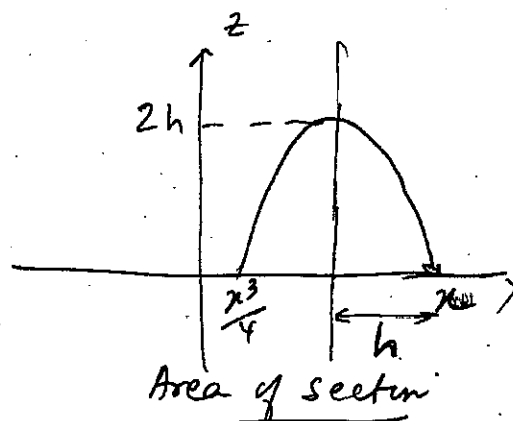
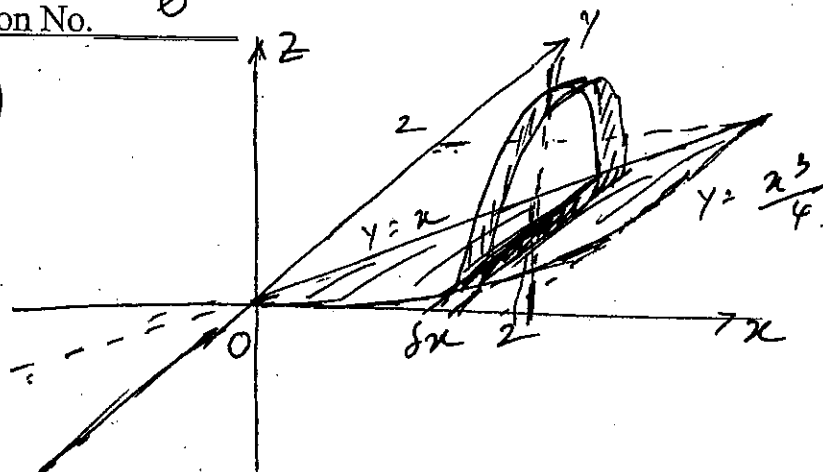
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Question No. 6

(b)



Let a section of thickness δx be taken $\perp$ to x -axis
 Using Simpson's rule, the area of the section is

$$A(x) = \frac{h}{3} (0 + 4(2h) + 0)$$

$$\text{where } h = \frac{x - x^3/4}{2} = \frac{4x - x^3}{8}$$

$$\begin{aligned} \therefore A(x) &= \frac{4x - x^3}{8 \times 3} \left(8 \cdot \frac{4x - x^3}{8} \right) \\ &= \frac{(4x - x^3)^2}{24} \end{aligned}$$

$$\delta V = A(x) \delta x$$

$$\therefore V = \int_0^2 \frac{(4x - x^3)^2}{24} dx$$

$$= \frac{1}{24} \int_0^2 (16x^2 - 8x^4 + x^6) dx$$

$$= \frac{1}{24} \left[\frac{16x^3}{3} - \frac{8x^5}{5} + \frac{x^7}{7} \right]_0^2 = \frac{1}{24} \left[\frac{16}{3}(8) - \frac{8}{5}(32) + \frac{1}{7}(128) - 0 \right]$$

$$= \frac{128}{315} \text{ u}^3$$



ANSWER SHEET

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Question No. 6

(c) $u_n = u_{n-1} + 6u_{n-2}$, $n \geq 3$, $u_1 = 1$, $u_2 = -12$

To prove

$$u_n = -6 \left[(-2)^{n-2} + 3^{n-2} \right]$$

Step 1 : prove true for $n=1$ and $n=2$

$$u_1 = -6 \left[(-2)^{-1} + 3^{-1} \right]$$

$$= -6 \left[-\frac{1}{2} + \frac{1}{3} \right]$$

$$= -6 \cdot \frac{-1}{6}$$

$$= 1 \text{ True}$$

$$u_2 = -6 \left[(-2)^0 + 3^0 \right]$$

$$= -6(1+1)$$

$$= -12 \text{ True}$$

Hence true for $n=1$ and $n=2$ Step 2 Assume true for $n=k$ & $n=k+1$, $k \geq 1$

ie $u_k = -6 \left[(-2)^{k-2} + 3^{k-2} \right]$, $u_{k+1} = -6 \left[(-2)^{k-1} + 3^{k-1} \right]$

Step 3 : prove true for $n=k+2$

ie. $u_{k+2} = -6 \left[(-2)^k + 3^k \right]$

$$\text{LHS} : u_{k+2} = u_{k+1} + 6u_k$$

$$= -6 \left[(-2)^{k-1} + 3^{k-1} \right] - 36 \left[(-2)^{k-2} + 3^{k-2} \right] \text{ from step 2}$$

$$= -6 \left[\frac{(-2)^k}{-2} + \frac{3^k}{3} \right] - 36 \left[\frac{(-2)^k}{4} + \frac{3^k}{9} \right]$$

$$= 3(-2)^k + 2(3^k) - 9(-2)^k - 4(3^k)$$

$$= -6(-2)^k - 6(3^k)$$

$$= -6 \left[(-2)^k + 3^k \right] = \text{RHS}$$

Hence true for $n=k+2$ Step 4 : By the second principle of mathematical induction, it is true for all $n \geq 1$



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Question No. 7

$$\begin{aligned}(a) (i) \cos 4\theta + i \sin 4\theta &= (\cos \theta + i \sin \theta)^4 \\&= \cos^4 \theta + 4i \cos^3 \theta \sin \theta + 6 \cos^2 \theta (i \sin \theta)^2 + 4 \cos \theta (i \sin \theta)^3 \\&\quad + (i \sin \theta)^4 \\&= \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta\end{aligned}$$

Equating real and imaginary parts.

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta \quad - (1)$$

$$\sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta \quad - (2)$$

$$(2) \div (1)$$

$$\tan 4\theta = \frac{4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta}{\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta}$$

Dividing each term on RHS by $\cos^4 \theta$

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$$



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Question No. 7

$$(a) \quad (ii) \quad x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$$

$$x^4 - 6x^2 + 1 = 4x - 4x^3$$

$$\text{or } \frac{4x - 4x^3}{1 - 6x^2 + x^4} = 1$$

$$\text{Let } x = \tan \theta$$

$$\text{then } \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta} = 1$$

Using the identity from part (i)

$$\tan 4\theta = 1$$

$$4\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$$

$$\therefore \theta = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16}$$

$$= \frac{\pi}{16}, \frac{5\pi}{16}, -\frac{7\pi}{16}, -\frac{3\pi}{16}$$

$$\therefore x = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, \tan(-\frac{7\pi}{16}), \tan(-\frac{3\pi}{16})$$
$$= \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, -\tan \frac{7\pi}{16}, -\tan \frac{3\pi}{16}$$

Product of roots = 1 from the polynomial equation

$$\therefore \tan \frac{\pi}{16} \times \tan \frac{5\pi}{16} \times -\tan \frac{7\pi}{16} \times -\tan \frac{3\pi}{16} = 1$$

$$\text{or } \tan \frac{\pi}{16} \times \tan \frac{5\pi}{16} \times \tan \frac{7\pi}{16} \times \tan \frac{3\pi}{16} = 1$$



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Question No. 7

$$(b) P(x) = x^4 + x^3 - 6x^2 - 4x + 48 = 0$$

Let the roots be $\alpha, \beta, \gamma, \delta$, let $\alpha\beta = 6$

$$\alpha + \beta + \gamma + \delta = -1 \quad \text{--- (1)}$$

$$\alpha\beta + \beta\gamma + \gamma\delta + \alpha\delta + \alpha\gamma + \beta\delta = -6 \quad \text{--- (2)}$$

$$\alpha\beta\gamma + \beta\gamma\delta + \gamma\delta\alpha + \alpha\beta\delta = 4 \quad \text{--- (3)}$$

$$\alpha\beta\gamma\delta = 48 \quad \text{--- (4)}$$

$$\gamma\delta = 48/6 = 8$$

$$P(x) = (x-\alpha)(x-\beta)(x-\gamma)(x-\delta) = 0$$

$$\Rightarrow (x^2 - (\alpha+\beta)x + \alpha\beta)(x^2 - (\gamma+\delta)x + \gamma\delta) = 0$$

$$\text{From (3)} \quad 6\gamma + 8\beta + 8\alpha + 6\delta = 4$$

$$\text{or } 8(\alpha+\beta) + 6(\gamma+\delta) = 4 \quad \text{--- (5)}$$

$$\text{(1) } \times 8 \quad 8(\alpha+\beta) + 8(\gamma+\delta) = -8 \quad \text{--- (6)}$$

$$\text{(5) } - \text{(6)}$$

$$-2(\gamma+\delta) = 12$$

$$\gamma+\delta = -6$$

$$\alpha+\beta = 5$$

$$\therefore P(x) = (x^2 - 5x + 6)(x^2 + 6x + 8) = 0$$

$$\text{or } (x-3)(x-2)(x+4)(x+2) = 0$$

$$\therefore x = 3, 2, -4, -2$$



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Question No. 7

(C) (1) $3x^4 - 4x^3 + m = 0$ ~~$\Rightarrow$~~

let $f(x) = 3x^4 - 4x^3 + m$

$f'(x) = 12x^3 - 12x^2$

$f''(x) = 36x^2 - 24x$

$f'(x) = 0 \quad 12x^2(x-1) = 0$ when $x = 1$

$x = 0$ or $x = 1$

$f''(x) \geq 0 \quad \therefore$ min. pt.

$f(0) = m, \quad f(1) = (3-4+m) = (m-1)$

x	0^-	0	0^+
$f'(x)$	$-$	0	$-$

say (i)

$\therefore x = 0$ is a horizontal poi.

$f''(x) = 0 \quad -$

For $f(x) = 0$ to have no real roots,

the poi and st. point must lie on the same side of x -axis i.e. above the x -axis ($\because y\text{-int.} > 0$, i.e. $m > 0$)

i.e. $f(0) \times f(1) \geq 0$

$(m) \times (m-1) > 0$

$\therefore m \neq 0$ or $m \geq 1$

$\therefore m > 1$ for no real roots

($\because$ y -int below x -axis

$\therefore$ there will be two intercepts.



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Question No. 7

(c) (ii) when $m=1$

$$f(0) = 1, f(1) = 0.$$

$\therefore$ st. point lies on the x -axis, point is above the x -axis

There is one real root



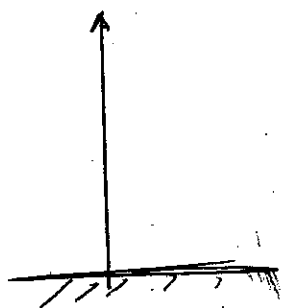
ANSWER SHEET

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Question No. 8

(a) (i)



$$\Sigma F = m\ddot{x} = -mg - mkv^2$$

$$\ddot{x} = \frac{v dv}{dx} = -g - kv^2$$

$$dx = - \frac{v dv}{g + kv^2}$$

$$x = \int - \frac{v dv}{g + kv^2}$$

$$= - \frac{1}{2k} \ln(g + kv^2) + C$$

when $x=0$, $v=u$

$$0 = - \frac{1}{2k} \ln(g + ku^2) + C$$

$$C = \frac{1}{2k} \ln(g + ku^2)$$

$$\therefore x = - \frac{1}{2k} \ln(g + kv^2) + \frac{1}{2k} \ln(g + ku^2)$$

$$= \frac{1}{2k} \ln \frac{g + ku^2}{g + kv^2}$$

for max H , $v=0$

$$H = \frac{1}{2k} \ln \frac{g + ku^2}{g}$$

$$\text{or } H = \frac{1}{2k} \ln \left(1 + \frac{ku^2}{g} \right)$$

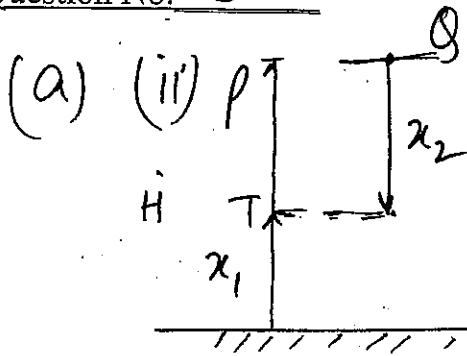


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Question No. 8



For Q: $\Sigma F = m\ddot{x} = mg - mkv^2$

for terminal velocity $\ddot{x} = 0$

$$g = kv^2$$
$$\text{or } v = \sqrt{g/k}$$

$$v \frac{dv}{dx} = g - kv^2$$

$$dx = \frac{v dv}{g - kv^2}$$

$$x = \int \frac{v dv}{g - kv^2}$$

$$= -\frac{1}{2k} \ln(g - kv^2) + C$$

when $x=0$, $v=0$

$$0 = -\frac{1}{2k} \ln g + C \quad \therefore C = \frac{1}{2k} \ln g$$

$$\therefore x = -\frac{1}{2k} \ln(g - kv^2) + \frac{1}{2k} \ln g$$

$$= \frac{1}{2k} \ln \frac{g}{g - kv^2}$$

Let the particles collide at T when for P, $x = x_1$, $v = v_1$,
and for Q, $x = x_2$, $v = v_2 \Rightarrow x_1 + x_2 = H$.

from (1)

$$x_1 = \frac{1}{2k} \ln \frac{g + kv_1^2}{g - kv_1^2} \quad x_2 = \frac{1}{2k} \ln \frac{g}{g - kv_2^2}$$



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Question No. 8

$$a)(ii) \quad x_1 + x_2 = H = \frac{1}{2k} \ln \left(1 + \frac{ku^2}{g} \right)$$

$$\frac{1}{2k} \ln \left(\frac{g+ku^2}{g} \right) = \frac{1}{2k} \ln \left(\frac{g+ku^2}{g+kv_1^2} \right) + \frac{1}{2k} \ln \left(\frac{g}{g+kv_2^2} \right)$$

$$\frac{g+ku^2}{g} = \frac{g+ku^2}{g+kv_1^2} \cdot \frac{g}{g+kv_2^2}$$

$$(g+kv_1^2)(g+kv_2^2) = g^2$$

$$k^2 \left(\frac{g}{k} + v_1^2 \right) \left(\frac{g}{k} + v_2^2 \right) = g^2$$

$$\left(\frac{g}{k} + v_1^2 \right) \left(\frac{g}{k} + v_2^2 \right) = \frac{g^2}{k^2}$$

$$g(v_1^2 + v_2^2) + g^2 = g^2$$

$$g(v_1^2 + v_2^2) = 0$$

$$v_1^2 + v_2^2 = 0$$

$$\text{Dividing by } v_1^2 v_2^2$$

$$\frac{1}{v_2^2} + \frac{1}{v_1^2} = \frac{1}{v_1^2 v_2^2}$$



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Question No. 8

$$(b) \quad x^2 + y^2 = xy + 3 \quad \text{--- (1)}$$

Differentiating implicitly with respect to x

$$2x + 2y \frac{dy}{dx} = x \frac{dy}{dx} + y$$

$$(x - 2y) \frac{dy}{dx} = 2x - y$$

$$\therefore \frac{dy}{dx} = \frac{2x - y}{x - 2y}$$

for stationary points $y' = 0$

$$2x - y = 0 \quad \therefore y = 2x \quad \text{--- (2)}$$

Solving (1) & (2) simultaneously, sub $y = 2x$ into (1)

$$x^2 + 4x^2 = 2x^2 + 3$$

$$3x^2 = 3 \quad x = 1, \quad y = 2$$

$$\text{or } x = -1, \quad y = -2$$

$$y'' = \frac{(x - 2y)(2 - y') - (2x - y)(1 - 2y')}{(x - 2y)^2}$$

(i) When $y' = 0, x = 1, y = 2$

$$y'' = \frac{(1 - 4)(2)}{(1 - 4)^3} < 0$$

 $\therefore (1, 2)$ is max pt(ii) When $y' = 0, x = -1, y = -2$

$$y'' = \frac{(-1 + 4)(2)}{(1 + 4)^3} > 0$$

 $\therefore (-1, -2)$ is min pt



ANSWER SHEET

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Question No. 8

(C) (i) For the branch $X - P - Q - Y$

Current will not flow if P defective or Q defective
or P and Q defective.

$\therefore P(\text{no current flow through } X - P - Q - Y)$

$$= p(1-p) + (1-p)p + p \cdot p$$

$$= p - p^2 + p - p^2 + p^2$$

$$= 2p - p^2 = p(2-p)$$

Similarly $P(\text{no current flow through } X - R - S - Y)$

$$= 2p - p^2 = p(2-p)$$

$$\therefore P(\text{no current flow from } X \text{ to } Y) = \frac{(2p - p^2)^2}{p^2(2-p)^2}$$

(ii) let q = probability of no current through any one
of the four circuits. then $q = (2p - p^2)^2 = p^2(2-p)^2$
then from above.

$$P(\text{no current from } A \text{ to } B) = \frac{(2q - q^2)^2}{q^2(2-q)^2} \text{ where } q = (2p - p^2)^2$$

$$\begin{aligned} \text{Probability} &= p^4(2-p)^4 \left[2 - p^2(2-p)^2 \right]^2 = p^4(2-p)^4 \left[2 - p^2(4-4p+p^2) \right]^2 \\ &= p^4(2-p)^4 (2-4p^2+4p^3-p^4)^2 \end{aligned}$$